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```
clc;
clear;
close all;

% EXPERIMENT 9: UNIFORM QUANTIZATION AND FINITE WORD-LENGTH EFFECTS
```

1. PARAMETERS

```
A = 1;
f = 50;
Fs = 1000;
N = 1000;

n = 0:N-1;
x = A*sin(2*pi*f*n/Fs);

B = 8;
bits = [4 8 12 16];
```

2. 8-BIT QUANTIZATION

```
D = 2*A/2^B;

xr = D*round(x/D);   % Rounding
xt = D*fix(x/D);     % Truncation

er = x-xr;           % Rounding error
et = x-xt;           % Truncation error

SQr = 10*log10(mean(x.^2)/mean(er.^2));
SQt = 10*log10(mean(x.^2)/mean(et.^2));
```

3. OUTPUT QUANTIZATION

```
y = x;
yq = D*round(y/D);
```

```
eo = y-yq;

SQout = 10*log10(mean(y.^2)/mean(eo.^2));
```

4. WORD-LENGTH COMPARISON

```
step = zeros(size(bits));
sqnrR = zeros(size(bits));
sqnrT = zeros(size(bits));
maxER = zeros(size(bits));
maxET = zeros(size(bits));

for k = 1:length(bits)
    b = bits(k);
    d = 2*A/2^b;

    qr = d*round(x/d);
    qt = d*fix(x/d);

    er1 = x-qr;
    et1 = x-qt;

    step(k) = d;

    sqnrR(k) = 10*log10(mean(x.^2)/mean(er1.^2));
    sqnrT(k) = 10*log10(mean(x.^2)/mean(et1.^2));

    maxER(k) = max(abs(er1));
    maxET(k) = max(abs(et1));
end
```

5. DISPLAY RESULTS

```
fprintf('\n===== QUANTIZATION RESULTS =====\n');
fprintf('Amplitude = %.1f\n',A);
fprintf('Frequency = %d Hz\n',f);
fprintf('Sampling Frequency = %d Hz\n',Fs);
fprintf('Number of Samples = %d\n',N);
fprintf('Quantization Bits = %d\n',B);
fprintf('8-bit Step Size = %.7f\n',D);
fprintf('Rounding SQNR = %.4f dB\n',SQr);
fprintf('Truncation SQNR = %.4f dB\n',SQt);
fprintf('Output SQNR = %.4f dB\n',SQout);
```

```
===== QUANTIZATION RESULTS =====
Amplitude = 1.0
Frequency = 50 Hz
Sampling Frequency = 1000 Hz
Number of Samples = 1000
Quantization Bits = 8
8-bit Step Size = 0.0078125
Rounding SQNR = 48.9339 dB
```

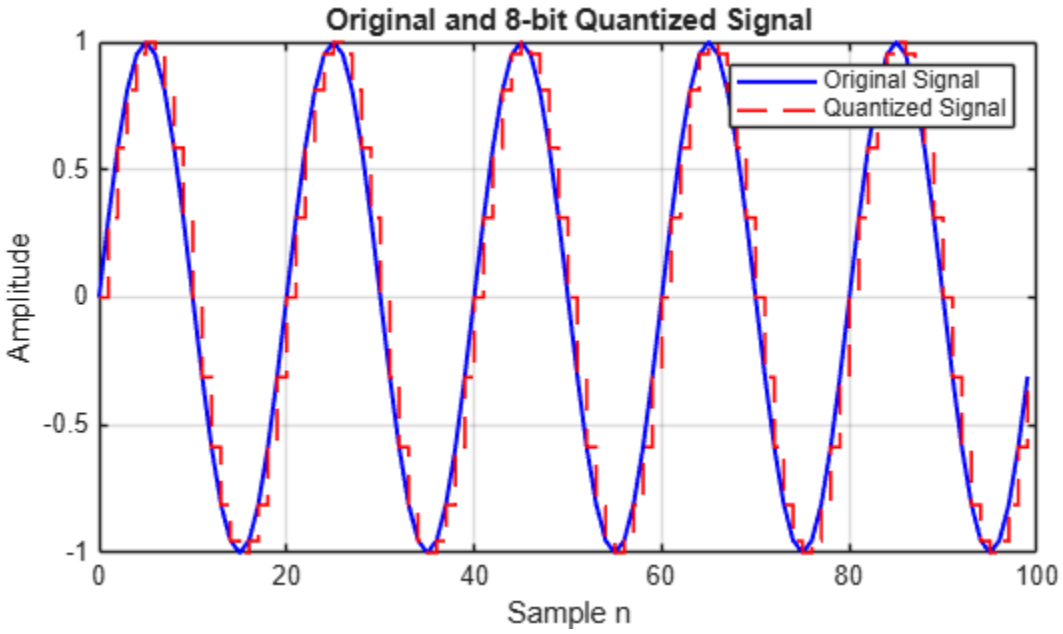
```
Truncation SQNR = 45.2931 dB
Output SQNR = 48.9339 dB
```

6. GRAPH 1: ORIGINAL AND QUANTIZED SIGNAL

```
figure(1);

plot(n(1:100),x(1:100),'b','LineWidth',1.5);
hold on;
stairs(n(1:100),xr(1:100),'r--','LineWidth',1.2);

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Original and 8-bit Quantized Signal');
legend('Original Signal','Quantized Signal');
```

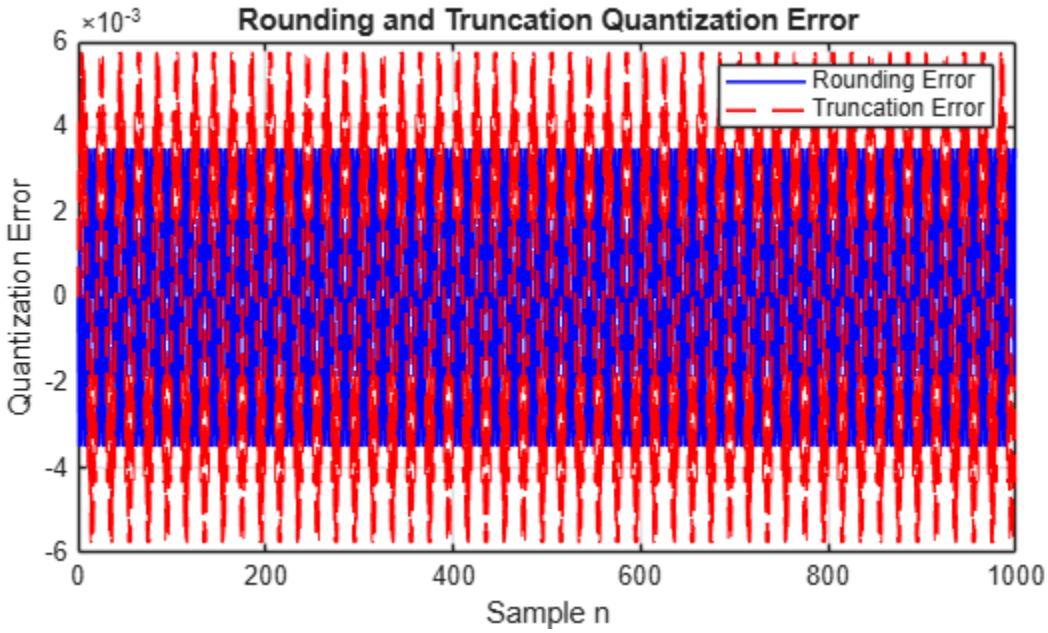


7. GRAPH 2: QUANTIZATION ERROR

```
figure(2);

plot(n,er,'b','LineWidth',1.2);
hold on;
plot(n,et,'r--','LineWidth',1.2);

grid on;
xlabel('Sample n');
ylabel('Quantization Error');
title('Rounding and Truncation Quantization Error');
legend('Rounding Error','Truncation Error');
```

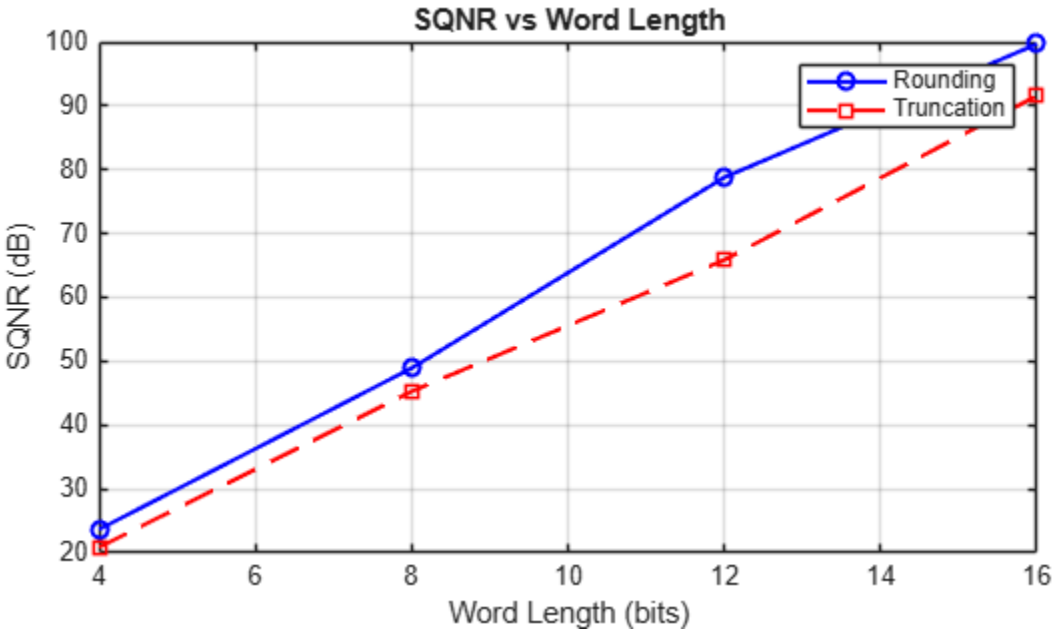


8. GRAPH 3: SQNR VS WORD LENGTH

```
figure(3);

plot(bits,sqnrR,'bo-','LineWidth',1.5);
hold on;
plot(bits,sqnrT,'rs--','LineWidth',1.5);

grid on;
xlabel('Word Length (bits)');
ylabel('SQNR (dB)');
title('SQNR vs Word Length');
legend('Rounding','Truncation');
```

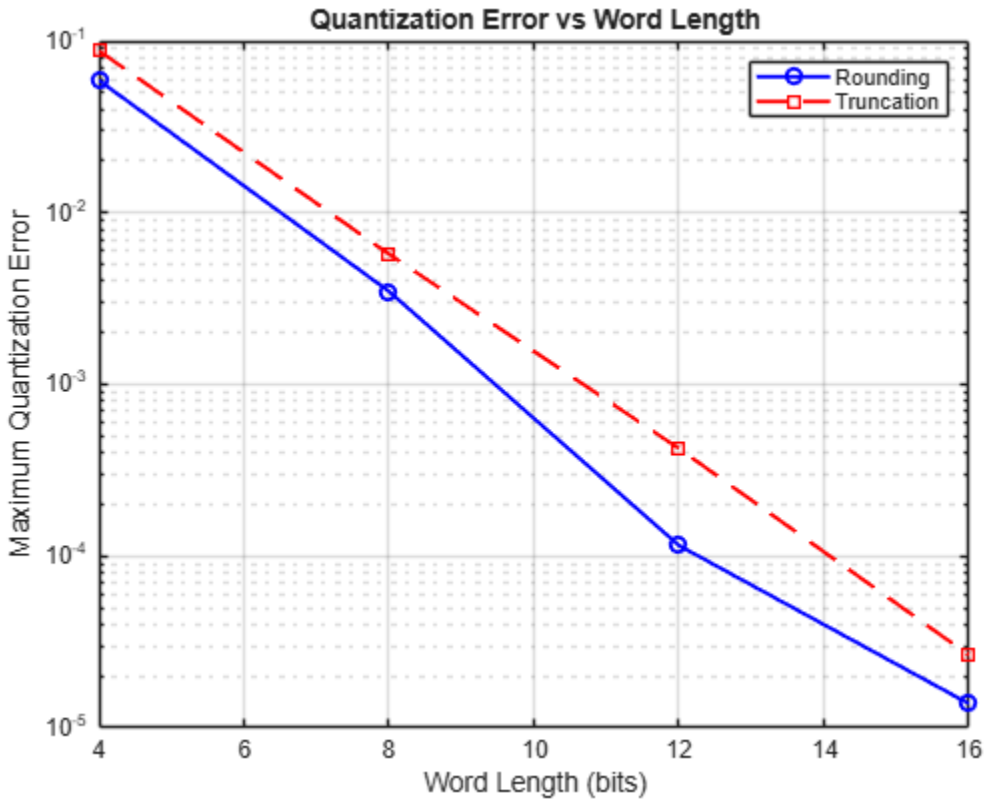


9. GRAPH 4: MAXIMUM ERROR VS WORD LENGTH

```
figure(4);

semilogy(bits,maxER,'bo-','LineWidth',1.5);
hold on;
semilogy(bits,maxET,'rs--','LineWidth',1.5);

grid on;
xlabel('Word Length (bits)');
ylabel('Maximum Quantization Error');
title('Quantization Error vs Word Length');
legend('Rounding','Truncation');
```



FINAL RESULT

```
fprintf('\n===== FINAL RESULT =====\n');
fprintf('Uniform quantization using rounding and truncation was
performed.\n');
fprintf('Quantization error and SQNR were calculated.\n');
fprintf('Increasing word length reduces quantization error and improves
SQNR.\n');
fprintf('Finite word-length effects are verified.\n');
```

```
===== FINAL RESULT =====
Uniform quantization using rounding and truncation was performed.
Quantization error and SQNR were calculated.
Increasing word length reduces quantization error and improves SQNR.
Finite word-length effects are verified.
```

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